# Matrix Theory II–III: Rank and Systems of Linear Equations

From the class book: Section 1.5 and Chapter 2, Sections 2.1 to 2.5

## 1. Key definitions

- A system of $m$ equations in $n$ unknowns over a field $F$ is $Ax = b$, with $A \in M_{m\times n}(F)$, $x \in F^n$, $b \in F^m$.
- $[A \mid b]$ is the **augmented matrix**, of size $m \times (n+1)$.
- The system is **homogeneous** if $b = 0$, **consistent** if it has a solution and **inconsistent** otherwise.
- Writing $A = [c_1 \cdots c_n]$ by columns, $Ax = x_1c_1 + \cdots + x_nc_n$. So $Ax = b$ is solvable exactly when $b$ lies in the **column space** of $A$.

**Null space and rank.** $N(A) = \{x : Ax = 0\}$ is a subspace of $F^n$. The **rank** of $A$ is the dimension of its column space (equal to that of its row space), the number of pivots in an echelon form, and the size of the largest non-zero minor. **Rank–nullity:** $\operatorname{rank} A + \dim N(A) = n$.

**Theorem (Structure of the solution set)**

- The solutions of $Ax = 0$ form $N(A)$, of dimension $n - \operatorname{rank} A$.
- If $x_p$ is one solution of $Ax = b$, all solutions are $x_p + N(A)$. This is a subspace only when $b = 0$.
- If $u, v$ solve $Ax = b$, so does $\lambda u + (1-\lambda)v$ for every scalar $\lambda$, and $u - v \in N(A)$.
- If $b \neq 0$, the system is consistent and $\operatorname{rank} A = r$, the largest number of linearly independent solutions is $n - r + 1$.

## 2. Main results

**Theorem (Rouché–Capelli consistency test)**

- $Ax = b$ is consistent iff $\operatorname{rank} A = \operatorname{rank}[A \mid b]$.
- In that case, with $r = \operatorname{rank} A$: the solution is unique iff $r = n$.
- If $r < n$ there are $n - r$ free variables: infinitely many solutions over an infinite field, exactly $q^{\,n-r}$ over $\mathbb{F}_q$.

| Condition | Homogeneous $Ax = 0$ | Non-homogeneous $Ax = b$ |
|---|---|---|
| $\operatorname{rank} A < \operatorname{rank}[A\|b]$ | impossible | no solution |
| $\operatorname{rank} A = \operatorname{rank}[A\|b] = n$ | only $x = 0$ | unique solution |
| $\operatorname{rank} A = \operatorname{rank}[A\|b] < n$ | infinitely many | infinitely many |
| $m < n$ | always a non-zero solution | none or infinitely many |
| $m = n,\ \det A \neq 0$ | only $x = 0$ | unique: $x = A^{-1}b$ |
| $m = n,\ \det A = 0$ | infinitely many | none or infinitely many |

**Important results: full rank and one-sided inverses**

- **Full row rank** ($\operatorname{rank} A = m \leq n$): $Ax = b$ is solvable for every $b$; $A$ has a right inverse ( $AB = I_m$). If $m < n$ the solution is never unique.
- **Full column rank** ($\operatorname{rank} A = n \leq m$): $Ax = b$ has at most one solution; $Ax = 0$ only has $x = 0$; $A$ has a left inverse, e.g. $(A^TA)^{-1}A^T$ over $\mathbb{R}$. If $m > n$, some $b$ give no solution.
- **Square** ($m = n$): $\det A \neq 0 \iff \operatorname{rank} A = n \iff Ax = 0$ has only $x = 0 \iff Ax = b$ is solvable for every $b \iff Ax = b$ has a unique solution for some $b$.
- $\{X \in M_{n\times p}(F) : AX = 0\}$ has dimension $p\,(n - \operatorname{rank} A)$, since each column of $X$ solves $Ax = 0$.

**Theorem (Rank inequalities)**

- $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$; multiplying by an invertible matrix does not change the rank.
- Sylvester: for $A$ ($m \times n$) and $B$ ($n \times p$), $\operatorname{rank} A + \operatorname{rank} B - n \leq \operatorname{rank}(AB)$.
- $\operatorname{rank}(A + B) \leq \operatorname{rank} A + \operatorname{rank} B$; over $\mathbb{R}$, $\operatorname{rank}(A^TA) = \operatorname{rank} A$.
- If $AB = 0$ then $\operatorname{rank} A + \operatorname{rank} B \leq n$.
- $\operatorname{rank}(\operatorname{adj} A)$ is $n$, $1$ or $0$ according as $\operatorname{rank} A$ is $n$, $n - 1$ or $\leq n - 2$.

**Theorem (Systems over subfields and finite fields)**

- If $K \subseteq L$ are fields and $A$ has entries in $K$, the rank of $A$ is the same over $K$ and over $L$.
- So if $Ax = b$ (entries in $K$) has a solution in $L^n$, it has one in $K^n$: rational systems with a complex solution have a rational solution. Integer solutions need not exist ($2x = 1$).
- Over $\mathbb{F}_p$, $A \in M_n(\mathbb{Z})$ is invertible iff $p \nmid \det A$. Reducing mod $p$ is not a subfield change, so the rank can drop.

## 3. Methods

**Method: solving or classifying a system.**

- (1) Write $[A \mid b]$ and reduce to row echelon form; row operations do not change the solution set.
- (2) A row $(0 \cdots 0 \mid c)$ with $c \neq 0$ means the system is inconsistent.
- (3) Otherwise count the pivots $r$. If $r = n$ the solution is unique; if $r < n$, express the $r$ basic variables in terms of the $n - r$ free ones.
- (4) For a square system compute $\det A$ first: if it is non-zero, the solution is unique and nothing else needs checking.

**Method: systems with a parameter $k$.**

- (1) Square system: solve $\det A(k) = 0$. For every other $k$ the solution is unique.
- (2) For each root $k_0$, substitute and row reduce $[A(k_0) \mid b]$; compare $\operatorname{rank} A$ with $\operatorname{rank}[A|b]$ to decide between no solution and infinitely many.
- (3) Non-square system: reduce with $k$ kept as a symbol and watch the last pivot; it is usually linear in $k$, and its vanishing changes the rank.

**Cramer's rule.**

- If $A$ is invertible, the unique solution has $x_j = \det A_j / \det A$, where $A_j$ is $A$ with column $j$ replaced by $b$.
- Use it when the question asks for one unknown only, or for its sign or size.

**Geometry in $\mathbb{R}^3$.**

- Each equation is a plane; three equations in three unknowns meet in a point (rank $3$), a line ( $\operatorname{rank} A = \operatorname{rank}[A|b] = 2$) or a plane (both ranks $1$).
- $\operatorname{rank} A = 2 < \operatorname{rank}[A|b]$: no common point (two parallel planes, or a triangular prism). $\operatorname{rank} A = 1 < \operatorname{rank}[A|b]$: parallel planes.
- Over $\mathbb{R}$ or $\mathbb{C}$ a system never has exactly $2$ (or any finite number above $1$) of solutions.

## 4. Exam shortcuts and common traps

- The number of free variables is $n - \operatorname{rank} A$: unknowns minus pivots.
- Look for a combination of equations that gives $0 = c \neq 0$ (for example the sum of all equations) before row reducing.
- For $kx + y + z = 1,\ x + ky + z = 1,\ x + y + kz = 1$: $\det = (k-1)^2(k+2)$; unique for $k \neq 1, -2$, infinitely many at $k = 1$, none at $k = -2$.
- A Vandermonde coefficient matrix in $a, b, c$ has determinant $(b-a)(c-a)(c-b)$: unique solution iff $a, b, c$ are distinct.

- $Ax = \lambda x$ has a non-zero solution iff $\lambda$ is an eigenvalue of $A$; for a circulant with first row $(a, b, c)$, the eigenvalues are $a + b\omega + c\omega^2$, $\omega^3 = 1$.
- **Traps:** "unique" needs $\operatorname{rank} A = n$ *and* consistency; "more unknowns than equations" forces infinitely many solutions only for homogeneous systems; the solution set of $Ax = b$, $b \neq 0$, is not a subspace, so do not give its "dimension" as a vector space; over $\mathbb{F}_q$ the answer is a count $q^{n-r}$, not "infinitely many".

## 5. Solved examples

**Example 1.** For which $k$ is $x + y + z = 1,\ x + 2y + 4z = k,\ x + 4y + 10z = k^2$ consistent? Solve it then.

*Solution*

- $R_2 - R_1$, $R_3 - R_1$: $(0, 1, 3 \mid k - 1)$, $(0, 3, 9 \mid k^2 - 1)$.
- $R_3 - 3R_2$: $(0, 0, 0 \mid k^2 - 3k + 2)$, and $k^2 - 3k + 2 = (k - 1)(k - 2)$.
- So the system is consistent only for $k = 1$ or $k = 2$, with rank $2$ and one free variable.
- $k = 1$: $y = -3z$, $x = 1 + 2z$. $k = 2$: $y = 1 - 3z$, $x = 2z$.

**Example 2.** Classify
$x_1 + x_2 + x_3 + x_4 = 4,\ x_1 + 2x_2 + 3x_3 + 4x_4 = 5,\ x_1 + 3x_2 + 5x_3 + kx_4 = m$.

*Solution*

- Row reduction gives $\begin{pmatrix} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 0 & k - 7 & m - 6 \end{pmatrix}$.
- $k \neq 7$: rank $3$, one free variable, infinitely many solutions.
- $k = 7,\ m = 6$: rank $2$, two free variables, infinitely many solutions.
- $k = 7,\ m \neq 6$: the last row reads $0 = m - 6 \neq 0$, no solution.

**Example 3.** $A$ is a real $5 \times 4$ matrix and the solutions of $Ax = (1, 2, 3, 4, 5)^T$ are $\{(1 + 2s,\ 2 + 3s,\ 3 + 4s,\ 4 + 5s)^T : s \in \mathbb{R}\}$. Find $\operatorname{rank} A$.

*Solution*

- The solution set is a line, so $\dim N(A) = 1$ (spanned by $(2, 3, 4, 5)$).
- Rank–nullity in $\mathbb{R}^4$: $\operatorname{rank} A = 4 - 1 = 3$.

**Example 4.** Let $A$ be $4 \times 6$ of rank $3$ and $b \neq 0$ with $Ax = b$ consistent. How many linearly independent solutions can $Ax = b$ have? What is $\dim\{X \in M_{6\times 2} : AX = 0\}$?

*Solution*

- $\dim N(A) = 3$ with basis $y_1, y_2, y_3$. Then $x_p,\ x_p + y_1,\ x_p + y_2,\ x_p + y_3$ are independent, and every solution lies in their $4$-dimensional span: the answer is $n - r + 1 = 4$.
- Each of the $2$ columns of $X$ lies in $N(A)$: dimension $2 \cdot 3 = 6$.

**Example 5.** Count the solutions of
$x + y + z + w = 0,\ x + 2y + 3z + 4w = 0,\ x + 4y + 4z + w = 0$ over $\mathbb{F}_5$.

*Solution*

- The first three columns form $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 4 \end{pmatrix}$ with determinant $-3 \equiv 2 \neq 0$ in $\mathbb{F}_5$.
- So the rank is $3$, there is $4 - 3 = 1$ free variable, and the number of solutions is $5^1 = 5$.

**Example 6.** Solve $x + y + z = 1,\ y + z + w = 0,\ x + w = 1$ over $\mathbb{F}_2$ and over $\mathbb{R}$.

*Solution*

- Over $\mathbb{F}_2$ the third equation is the sum of the first two: rank $2$, consistent, $2^{4-2} = 4$ solutions.
- Over $\mathbb{R}$, $R_3 - R_1 = (0, -1, -1, 1 \mid 0)$; adding $R_2$ gives $2w = 0$. Rank $3$, so $w = 0$, $x = 1$, $y = -z$: infinitely many solutions.
- Moral: reducing modulo $p$ can lower the rank.

**Example 7.** For how many primes $p \leq 20$ does $Ax = b$, $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$, have a unique solution mod $p$ for every $b$?

*Solution*

- $\det A = (2-1)(3-1)(3-2) = 2$, so $A$ is invertible mod $p$ iff $p \neq 2$.
- The primes $3, 5, 7, 11, 13, 17, 19$ work: $7$ primes.